

Quick Check

Lesson 5: From Fertilisation to Birth: How Hormones and the Placenta Sustain Pregnancy

9 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.1: Reproduction in Animals | Lesson 5

A cow carries one calf for about 9 months. How does the developing calf get its oxygen and food while inside the mother?

- A** The calf swallows food through its mouth and breathes air through its nose inside the uterus
- B** The mother's blood flows directly into the calf's blood vessels and mixes with it
- C** The calf uses only the yolk stored inside its body until birth
- D** Substances diffuse across the thin membranes of the placenta between the mother's blood and the calf's blood

Which statement about IVF babies is correct?

- A** The embryo is placed in the mother's uterus and still develops inside her body
- B** The baby grows wholly in a test tube until birth
- C** The baby grows inside the mother only for the first week and then in a machine
- D** The baby does not need a placenta because it started in a laboratory

In humans, where does fertilisation normally take place, and where does the embryo then implant?

- A** Fertilisation in the oviduct; implantation in the uterine wall
- B** Fertilisation in the vagina; implantation in the cervix
- C** Fertilisation in the uterus; implantation in the oviduct
- D** Fertilisation in the ovary; implantation in the placenta

A learner draws a placenta and marks where oxygen and nutrients pass from the mother to the baby. Which labelled part is the site where this exchange happens?

- A** The placenta, across thin membranes
- B** The wall of the vagina
- C** The amniotic fluid, where the baby absorbs food
- D** The umbilical cord, where maternal blood mixes with fetal blood

Why does the mother's blood not mix with the baby's blood in the placenta, even though substances pass between them?

- A** They have separate blood supplies, and substances move by diffusion across thin membranes
- B** The umbilical cord pumps the mother's blood into the baby
- C** The mother's red blood cells are too small to enter the baby's blood
- D** Blood mixes only during labour, not during pregnancy

After implantation, which hormone from the embryo keeps the corpus luteum active so that progesterone continues to maintain the uterine lining?

A Oestrogen in late pregnancy

B Prolactin

C Oxytocin

D hCG

At the end of pregnancy, oxytocin causes the uterus to contract. These contractions push the baby against the cervix, which causes more oxytocin to be released. What is this type of control called?

- A** Hormone mixing
- B** Positive feedback
- C** Negative feedback
- D** Homeostasis

A frog releases thousands of eggs in a pond, while a cow produces one calf at a time. Which statement best explains this difference?

- A** The cow's calf is protected and nourished inside the mother and cared for after birth, so fewer offspring are needed
- B** Frog eggs have thick shells, so many are needed to break through them
- C** The frog's eggs are fertilised inside the body, so many are needed
- D** The cow has fewer eggs because cows are larger than frogs

Which pair correctly matches a hormone with its role in pregnancy, birth or lactation?

- A** Oxytocin: stops the uterine lining from forming
- B** Prolactin: supports milk production after birth
- C** Progesterone: causes labour contractions
- D** hCG: produced by the mother's breasts to make milk

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).